BEHIND

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Abstract

The goal of this research is to look into healthcare access disparities that lead to delayed and unmet healthcare needs among the elderly, as well as health inequality and healthcare cost burden. Any project's success is contingent on high-quality steps. We began by looking for the source of the problem. Our issue is that most elderly people do not have access to health care. For example, when my grandfather(Nour El-Deen Ezzat) suffered a stroke and died as a result of it, he was alone in the house, and no one came to his aid. After that, we looked for previous solutions. We observed a variety of devices, but smartwatches are the most popular today. It's nice and has many advantages, such as being tiny and pleasant for the patient, but it has some downsides. It's good, but it's better for athletes than for the elderly. When it comes to smartwatches, the elder should do it on his own, but dealing with technology is difficult for an elder. We began to decide our answer after seeing this and other solutions. We created three systems for our solution: one for the home, one for the patient, and one for us to sit in. the heart of two systems. The first system we made to measure house validity like temperature, gasses, etc. For the second system, we made it to monitor the patient's vital condition like heart rate, oxygen rate, and temperature of him to see the effort he does. Also we're making a face tracker & emotion set using an esp32 camera to detect emotions of a patient and if one is trying to rob him. In terms of the smartphone app, First, when the project is implemented on the ground, it will be linked to the hospital, police, pharmacy, and the patient's guardian. First, we created a feeding system for diabetics and anemia patients. In addition, we are creating a large number of conditions in order to diagnose the patient's condition. Finally, I'd like to show that we intend to create a comprehensive system for seniors and patients.

Introduction

Acronyms and abbreviations:

GDP: Gross Domestic Product

IoT: Internet of Things

DHT11: Humidity & Temperature sensor

LED: Light Emitting Diode

ECG: Electrocardiogram

W001A: Body temperature sensor

3D: Three Dimension

APA: American Psychological Association

The process of industrialization in Egypt depends on utilizing technology, natural resources, and labor. Egypt is a country with very limited natural resources. Most of manufacturing technologies are imported. Accordingly, in this research we are working to eradicate public health dereliction by investing technology to diminish the traditional techniques that are used in the medical appliances industry. We wish we could exploit Egypt's accessible resources to construct an intellectual Apparatus for monitoring patients' health based on the Internet of Things (IoT) property. That will make an immense progress in the medical equipment industry. The Patient Monitoring System (PMS) is a very critical monitoring systems, it is used for monitoring physiological signals including Electrocardiograph (ECG), Patient's environmental conditions -DHT11 Sensor - from temperature to humidity, Body Temperature and other Gases etc. In PMS, the multiple sensor and electrodes is used for receiving physiological signals such as ECG Electrodes, DHT11 and Temperature Probe to measure the physiological and environmental signals for. During treatment, it is highly important to continuously monitor the vital physiological signs of the patient. Therefore, patient monitoring systems has always been occupying a very important position in the field of medical devices. The continuous improvement of technologies not only helps us transmit the vital physiological signs to the medical personnel but also simplifies the measurement and as a result raises the monitoring efficiency of patients. we considered the efficiency, cost, time and eco-conservation as the most important design requirements needed in a well-developed PM device and constructed a testable prototype achieved the desired results which leads to a perfect monitoring device suits the Egyptian status

As a result, we follow the research methods of solving all healthcare issues that depends on most significant immigration in 21th century that's was the "Technology "and how to make internet of relation between Technology and healthcare under the title of "Improve the science and technology environment for all" We have invented and use a lot of technology to great effect in our lives. However, each piece of scientific advancement carries a hidden cost, quite often an environmental danger. This lesson explores some of these hidden dangers. Technology: A Two-Sided Coin Look around you in the room you're in. It's full of technology: from technology that has built its walls, to the screen you are looking at, to the chair you sit on. All of that was born out of human scientific and technological innovation. Our technological advancements have almost always been two-sided. They have helped us in one way but hurt us in another. We need in this time to improve the ways of technology too sole the Environmental problem. We do not have in Egypt Genetics. or heredity lab so we need all of this. Some of the negative impact of the technology is mainly on the environment such as pollution and destruction of the ecosystem. The lack of technology as in Egypt leads to the deterioration of other fields such as health, industry and commercial.

Healthcare and medical industry:

Health Care is simply caring for people and their lives. It is the art of delivering hope and attention in an active process that requires communication, collaboration, and decision-making across care providers and care settings. The health system in Egypt incorporates both the public and private sectors of the health insurance market. Health care services are provided by three main sectors, the government, the public sector, and the private sector. Although these sectors complement each other, the government is considered the principal provider of health care services in terms of capacity, expenditures and the different kinds of services offered. Approximately 50% of the population is covered by basic government health insurance, a further 30% is enrolled in private health insurance plans and 20% of Egyptians have no health insurance.

Health Services:

The State works on ensuring that all citizens get equal access to the integrated health care services as follows:

(1) Medical Services are provided by medical units affiliated to the Ministry of Health, other ministries and the private sector. These data are tabulated according to the number of hospitals, medical units, beds and pharmacies in each governorate.

(2) Production of vital drugs and vaccines.

(3) Treatment of citizens inside and outside Egypt, at the government's expense.

(4) Number of physicians, pharmacists and nursing staff belonging to the Ministry of Health, in addition to number of employees working at the Ministry of Health.

The industry is generally divided into two sectors: healthcare equipment and services; and pharmaceutical, biotechnology and other related life sciences.

The first sector of the healthcare industry, healthcare equipment and services, is comprised of companies that provide medical equipment, medical supplies, and health care, such as hospitals, home health care providers, private medical practices and nursing homes.

The second industry sector is comprised of companies that produce biotechnology, pharmaceuticals, and miscellaneous scientific services.

Different services in healthcare and medical industry:

The health-care industry provides diagnostic, healing, rehabilitation, and preventive services.

Diagnostic services:

Procedures that are used to determine the cause of an illness or disorder. Activities related to the diagnosis made by a physician or nurse practitioner, which may also be performed by nurses or other health professionals. There are generally three levels to diagnostic services. At the basic level, diagnostic procedures observe symptoms or use a simple microscope. At the next level, diagnostic services require equipment such as X-rays or culture facilities. At the more complex level, diagnostic services demand sophisticated equipment such as endoscopes and CT scan machines.

Problems associated with healthcare and medical industry:

1. Buildings and facilities are very old and in dire need of renovation and expansion due to the overload. Periodic maintenance is still an absent concept in public hospitals.

2. Doctors incredibility.

3. Shortage of nurses critically affects the operations in-state hospitals.

4. Lines are always endless and patients wait long hours for their consultation.

5. Modern technologies are still a distance away in the field of operation management.

6. The limited availability of equipped baby incubators and ICU s: According to the official count issued by the Ministry of Health, 2,382,000 births were recorded in 2010. Around 10% of the total count of births requires incubators while the total number of incubators in public hospitals and health facilities is 3,689. Hundreds of babies are denied the access of the only available 30 incubators in a typical public hospital. Same problem is found in the case of Intensive Care Units (ICU); the shortage results in loss of life. Patients in critical condition often have to be transferred to other hospitals because of lack of space in the ICU. These patients are barely hanging on to life and at least half of them die as they are being transferred. Those who don't die end up having serious complications because of the delay.

7. Over-crowdedness of public hospitals: Total number of public and central hospitals in Egypt is 263 hospitals serving a total population of around 91 million. The patient load is massive compared to the available facilities.

Hypothesis

We decided to work on the bio devices as its prices raised up recently and it is very complicate, it has huge size and such the bio innovation aren't found in Egypt, so we aimed to invest the available resources in Egypt and make bio devices that satisfies with the Egyptian economy and finance. We merged some ideas of huge devices (ECG (electrocardiography), hydrometer and thermometer) in such wearable device watch-like consists of a collection of sensors differ in their function and their composition.

Old ideas

The old devices were used to measure the rate, internal human temperature and the surrounding humidity and temperature are ECG, hydrometer and thermometer not considered oximeter or diabetes meter. **ECG (electrocardiography)**

The electrocardiogram or ECG (sometimes called EKG) is today used worldwide as a relatively simple way of diagnosing heart conditions. An electrocardiogram is a recording of the small electric waves being generated during heart activity. ECG is a biomedical device that measures the heart rate (pulse) from electric signals comes from the heart directly by connecting some wires include electrodes to the body of the patient and then send these signals to the monitor where it appears and graphed. The electric currents in the heart have been measured for more than a hundred years, but the fundamental function of the ECG as we know it today was developed by the Dutch scientist Willem Einthoven in the beginning of the 20th century. In 1924 Einthoven was awarded the Nobel Prize in Physiology or Medicine "... for his discovery of the mechanism of the electrocardiogram."

The mechanism of ECG:

• A normal heartbeat is initiated by a small pulse of electric current. This tiny electric "shock" spreads rapidly in the heart and makes the heart muscle contract. In the heart, there are cells specialized in producing electricity. These are called pacemaker cells. They produce electricity by quickly changing their electrical charge from positive to negative and back.

• The first electric wave in a heartbeat is initiated at the top of the heart. Because of the heart muscle cell's ability to "spread" its electric charge to adjacent heart muscle cells, this initial wave will be enough to start a chain reaction.

• Wires ends with electrodes are connecting to the body to the chest near the heat. Making the electrodes sensitive enough was a challenge in the early days of the ECG. In the late 1800's early attempts to measure the electric activity in frog's hearts were successful only when the hearts were exposed directly to the measuring equipment. The measuring conditions were indeed difficult; the scientists wanted to be able to measure the electric signals without having to enter inside the body. The problem was that the electric wave got weaker since it had to travel through bone and body tissue before reaching an electrode applied on the skin. This problem was solved a couple of decades later by Willem Einthoven. He managed to improve the sensitivity of the ECG by using a string galvanometer.

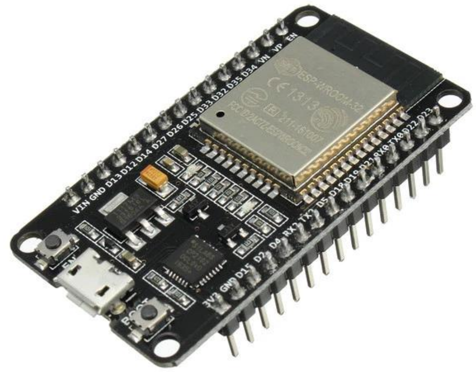
• The electric waves in the heart are recorded in millivolts by the electrocardiograph. The waves are registered by electrodes placed on certain parts of the body. Each electrode controls an ink needle that writes on a grid paper. The higher the intensity of the electric wave, the higher up the needle will move on the paper. The paper moves at a certain speed beneath the needle, resulting in an ink curve. • An ECG curve has different characteristics depending on the location of the electrode recording it.

• When the curve falls below the base line it shows a negative deflection and when it rises above the base line it is a result of positive deflection. A negative deflection indicates that the recorded wave has traveled away from the electrode and a positive deflection means it has traveled towards it • When the heart muscle is at rest, the pacemaker cells are negatively charged and when the heart contracts they are positively charged. When a positive wave is recorded by a positive electrode, the ECG curve will be pointing upwards and vice versa

Background Research

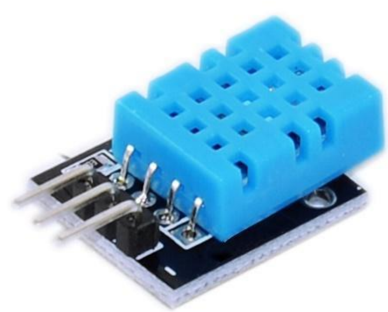
very Successful project depends on high qualified and documented scientific base and the hypothesis of our project considered in the concept of applying the IOT in health care , A healing experience starts with a quality welcome followed by a smooth admission enabled by easy onboarding using secure Wi-Fi ending with a user-friendly room phone and virtual doctor on your phone . All are essential for maintaining a link with the outside world during a hospital stay. Patient digital engagement continues within the facility with self-navigation services and interactions with the hospital and their care team via Rainbow, during their stay and after discharge. To improve care delivery, clinical staff expect full mobility and a high performing WLAN to connect a variety of IoMT, medical equipment and devices on the move for ubiquitous access to health data. They need to simplify their daily work using tools to communicate and collaborate easily with their peers, within and beyond the ward, with nurses, and with patients. An asset tracking solution reduces time to locate people and equipment, improving efficiency and enabling more patient care time. Devices in the form of wearables like fitness bands and other wirelessly connected devices like blood pressure and heart rate monitoring cuffs, glucometer etc. give patients access to personalized attention. These devices can be tuned to remind calorie count, exercise check, appointments, blood pressure variations and much more.

Materials and Methods



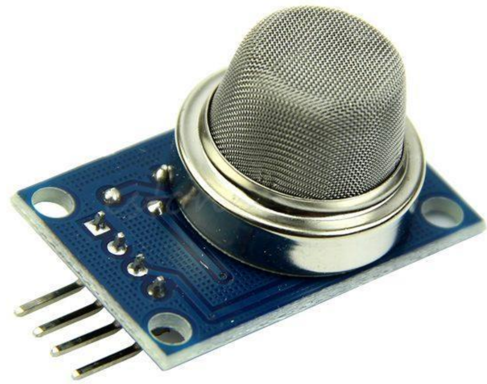
For materials First we used for box and hand ESP32 Board

We used it to control the two systems and it cost us 250 for each.



We used DHT11 to measure the temperature and humidity of room and it costs us about 35 L.E

We used an MQ-5 Gas sensor to measure the CO2 and it cost about 40 L.E.



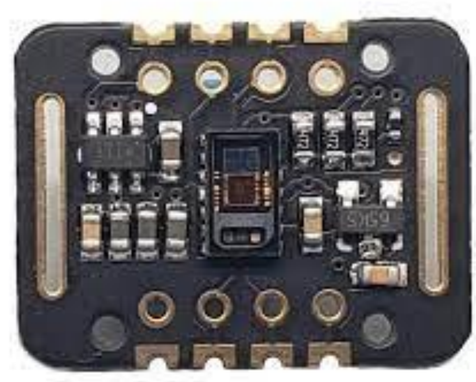
We used an MQ-7 Gas sensor to measure the butane and propane, it cost 40 L.E.

We used about 3 bread boards to put the jumpers in it and make the wire connection easier, we used 2 breadboard and cost us about 25L.E for each

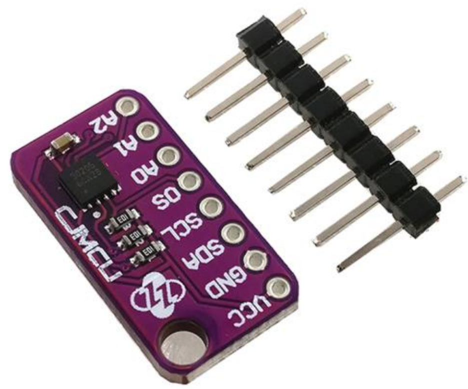
We used this camera to make the face detection of the patient and it costs about 350 L.E

We used Li-NH batteries because it’s re-chargeable and have a high current and good volt. We used 4 batteries and it costs us about 70 L.E

We used an MAX30102 sensor to measure Heart rate & Oxygen and it cost 200 L.E.



We used an MAX30205 sensor to measure Blood’s tempreature and it cost 600 L.E.



We use two servo motors to control the movement of the camera and it costs us 60 L.E.

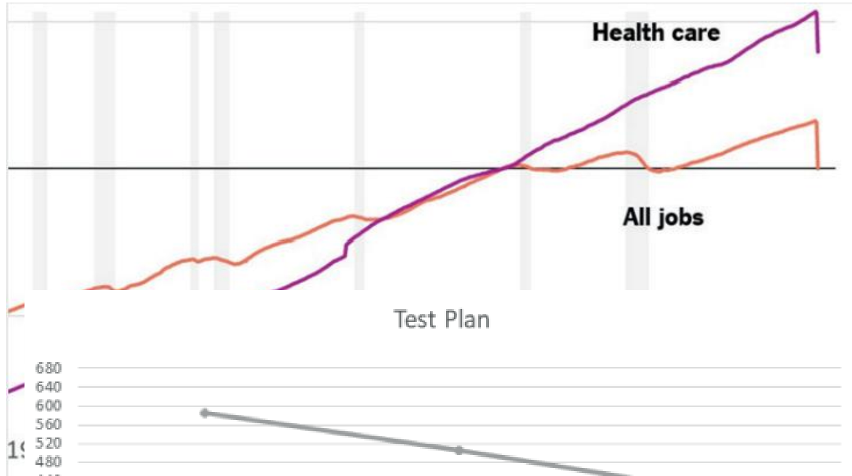
We used the LCD with I2C to show the data on it and it costs us about 60 L.E

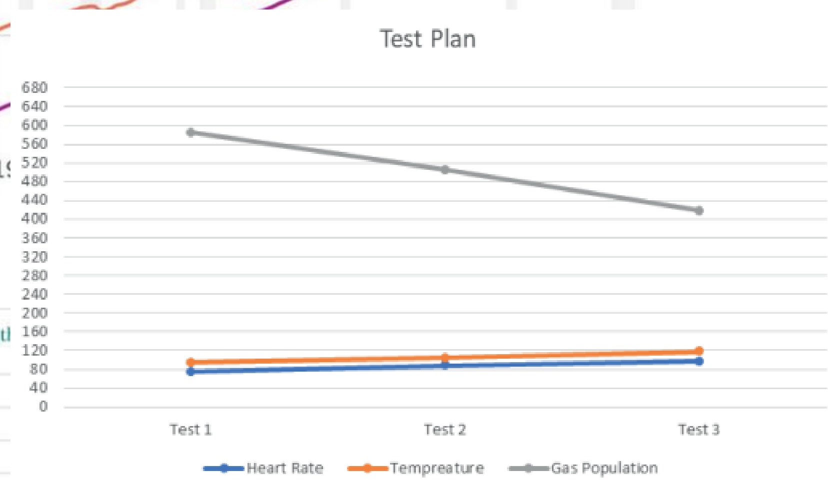
Methods

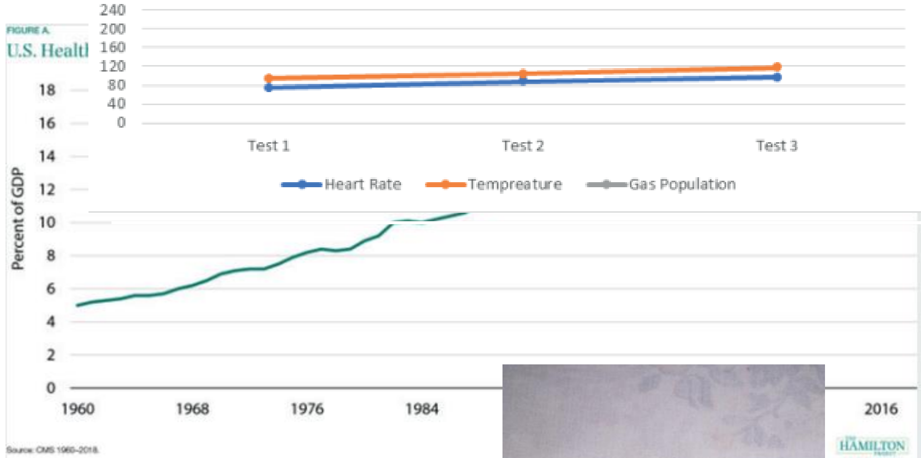
Our project consisted of three systems, as we previously stated. We started by making a box for home health. We went to the fab lab and used the coreldraw application to create a design, after which we bought acleric and cut it out with our design. We used the temperature and humidity of the patient's room (using DHT11) as well as any CO2 leaks in the room (Using MQ-5) and any butane or propane (Using MQ-7). We brought a camera with us, and Ai made it work. It monitors the patient's face and determines his status, which is the first new feature we've included because health care services don't just rely on human health; we also need to offer him the best possible environment to live in. We designed a glove for his health that analyses his heart rate, oxygen rate (using a pulse sensor), and sends all of this information to a doctor who follows up with the patient. If there is any deficiency in these things, the doctor will be notified, and he will take action right away. The third system is the mobile application, which, as I previously stated, serves a specific purpose. We built it using the flutter framework. Also we make wireless communication between gloves and the box using ESP-NOW protocol (it’s a protocol built inESP32).

Results and Discussion

As a result, a well-functioning healthcare system is a precondition for a well-functioning economy. Figure 12 depicts the ratio of increase in GDP of health care; GDP stands for Gross Domestic Product and refers to the increase in the amount of money we pay for something; as the graph shows, this issue has gotten worse in recent years; however, the project we are working on will address this issue; any successful project requires several factors, one of which is to address economic issues; therefore, we worked on this point because, in the long run, the GDP of healthcare will decrease due to increased economic expenditures; therefore, we worked on this. There must be a balance with jobs required; no job should be required more than others. Figure 1 shows that healthcare jobs are in high demand now, but there aren't enough doctors to go around. Our solution will address this issue; in the long 0run, if our solution is implemented, the demand for doctors will decrease; as a result, the load on doctors will decrease, which will benefit other jobs and restore balance for the required job.







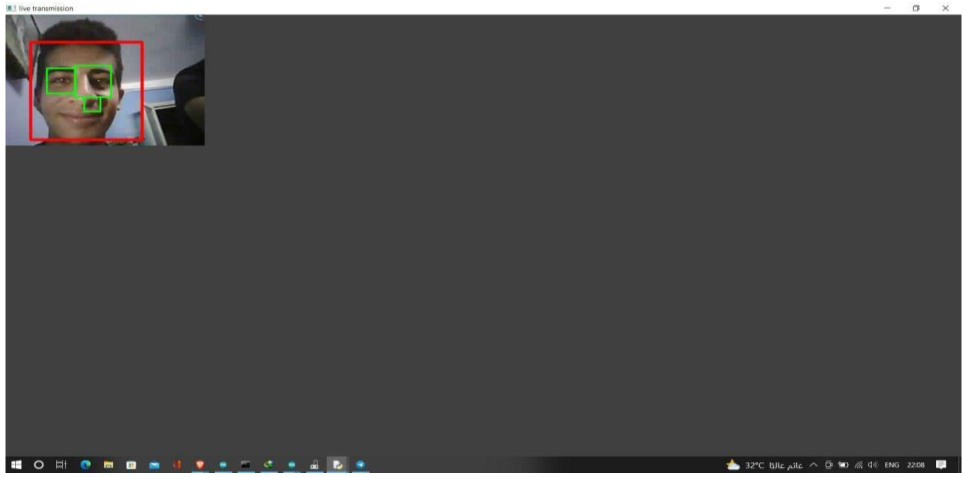
Now that we've finished, and since it's only a prototype, the cost won't exceed 1600 L.E., the project will have a lot of features, but we finally did a small test with the prototype, and it worked as expected. Sure, this solution isn't perfect, but we try our hardest to make the best, and this solution will make it easier for doctors all over the world, and it's almost as if a robot is working behind the scenes. Take care of yourself, your health, and your life. In conclusion, I'd want to show that the tests yielded faultless findings. prototype We can't get enough of it; the prototype will include additional functions, and our ambition is to. make it a global problem for the world.

Indeed, after exclusive research and several experiments to test the efficiency of the prototype, whole parts can be worked effectively and let me be more detailed and give a view of every individual system and the results and then the integrated brain system related with each other.

Firstly, the home system after giving all data from MQ-7 and DHT11 and send it to the LCD to give the server an alert to get the information of the hand to upload it on the server to get it by mobile application as shown in (Fig.1)

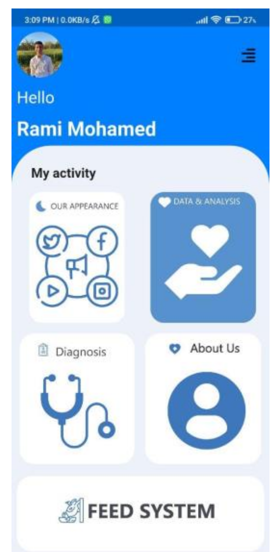
Secondly, Artifical inteligiouns toutaly acheiveid in the ESP32-CAM that allows to track the situation of the elders and evaluate the unexprected comas and regonition the expresition of the elders to maintain the mental health and comas that could lead to death and obvious preserving the situation and spread awearness to mental health mindset by this prevention as shown in (Fig.n)

After all the programming process with integrated and sophisticated packages of Flutter languach to achieve total bluetooth connection with mobile application.inaddition, Behind U connects elders to a board-certified doctor 24/7 through the convenience of phone or video visits from gathered data.Although mobile healthcare apps are intended to improve the quality of care for patients, the majority of them fail to match their expectations. According to the app usage data, most of the healthcare apps have less than 10,000 downloads. This is an extraordinarily low number in comparison to all of the mobile healthcare apps available for download. This app provides patients with quick and easy communication with doctors who are ready to listen and resolve their issues. nevertheless, the police,



embiolance, pharmacy and firefighters organization to deliver an action towards any errors.

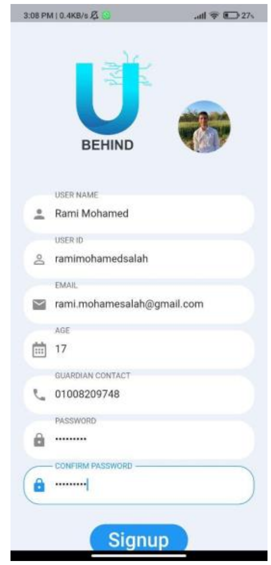
The healthcare app needs to provide clear and actionable information for all users. What we mean by this is that many apps fail to take patients with disabilities into consideration. It is important to cater to the needs of all potential users and provide reasonable and helpful functions.



Patients usually download healthcare apps to learn more about their condition, the causes and take action based on the available opportunities. It's important that apps provide interactive tools to keep users engaged and feel that they are receiving individualized care.

According to the Journal of Medical Internet Research, social features in healthcare apps enhanced patients' ability to commit to treatment plans and healthier habits. Online communities provide a space to interact with others going through similar situations. Fostering a positive and helpful community for end-users will yield a great return on investment for all.

Social features in healthcare apps allow users to connect and share messages of hope. This space is essential for those who have received an upsetting medical diagnosis and engaging with others can provide them with inspiration and strength during their road to recovery.



Design Requirements& Discussion

As we Know, Successful research program consists of the design requirements to follow this and to but you leg on the Right way of trusted scientific research Design requirements apparent the important characteristics that our solution must meet in order to be successful. Selecting the Design Requirements is a central process to successfully define our solution. As we know that not every solution is qualified to be applied and not every solution is feasible and may not satisfies with the current development but a successful solution passes the test of acceptance and accomplished the design requirement that help the solution to survive so we suggested our solution to be successful by accomplishing the desired design requirements by making a device that have the ability to measure the pulse of the patient, surrounding environmental conditions (humidity, temperature) also and the internal temperature of the patient. That device considers as an alternating solution which merge several devices in one device like (ECG (electrocardiography) that is responsible for calculating human heart rate, thermometer, hydrometer that is responsible for measuring humidity). This device accomplishes the desired design requirements that satisfies with the human and science desires.

Design requirements of our solution

- Cost
- Efficiency
- Eco-friendly
- Feasibility
- Safety
- Compact and Simplicity

1. High efficiency:

Efficiency is the quality of being efficient, for our solution to be accurate in the data provided from the Temperature and Humidity sensor, Body Temperature sensor and Pulse sensor. We used types of sensors that are most efficient in the medical field in relative with its price and its compatibility with the Arduino Uno shield. And as known that utilizing sensors is more sufficient and accurate than working by approximation as sensors have sensitivity to parts of millions or even billions that give precise result and this is actually what we concentrate on in our project as we utilized LM35 temperature sensor that measures the temperature accurately in addition we used humidity sensor and Arduino and by merging these sensors together we obtained the desired result that has the highest efficiency.

2. Low cost and Availability of the material:

According to this aspect, our solution consists of one deice that has supreme trait by merging all the properties of several devices like thermometer, hydrometer and ECG (electrocardiography) .These devices are responsible for many distinguished jobs that keep the patient under observation where it can measure the pulse (heart rate) by the ECG device , measuring the temperature of the patient body, temperature of the surrounding environment using the thermometer , calculating the surrounding humidity by the hydro meter so unleash your imagination for such like this device that merge all the properties of the other three massive devices in a collection of sensors not reach size of small box , so we have saved huge cost by making this device as an alternating for the other huge three devices. If we added up the price of device contains sensors nearly 25$ according to 350$ at least for the other three devices and this is truly enormous provide. We could save about 93% of the major price of the whole process and this indicates passing our solution the test of cost as it broke all the standard numbers in saving money with high efficiency. In brief, in making our prototype we considered using materials with the highest possible quality and cheapest price. Also, the materials should exist in Egypt and available in the Egyptian markets in addition to the availability of the required methods involved in our solution. We calculated the total cost of the prototype and compared it with other devices available in the market.

3. Environmentally friendly:

One of the important characteristics for the solution to apply it is the environmental impacts of this solution. So we made sure that our solution should have no negative impacts on the environment and that it is made from materials that doesn't harm the environment. We used electronic devices that are not polluted to the environment and having not terrible impact on it as it doesn't depend on consumption or any materials or chemicals that cause environmental hazards.

4. Compact and Simplicity:

Our prototype had to be easy to be used by any un-qualified person and also it has to be small in size so it is easier to be handled and more portable. So, it can be used by anyone and can be located in any place. Making it connected with the internet makes the data available for who it may concerns anywhere.

5. Applicability and Testability:

Applicability means that our solution will be applicable so that our country can apply and take use of its advantages and its products. And our solution will be applicable when its components that form it are available to be got. To make a prototype that is testable so that we can use it to get the results of the tests to see that if we achieved our design requirements or not. This is the most require part I nanny solution for any problem is to be applied in the real life and could be tested and this design requirement is based on two main parts, the first is the validity of the materials that construct the solution and the second on the economy that must satisfies with the process maintained by the solution, so if these required two things are accomplished or are available so it is consider to be feasible and this is exactly what our solution have and we could test it and conclude the results.

6. Safety:

AS I said before that our solution doesn’t contains any consumption or any reactions of chemicals and its simple that cannot cause electric chokes for any person who use this device.

Discussion

In brief, our prototype achieved the desired design requirements of

- cost
- efficiency
- safety
- compact & simplicity.

This device offers a good care for any patient that needs health care especially old people. The results of the tests we made on our prototype proved that it is a feasible, cheap, portable, accurate and simple device with an efficiency of 98.4% of the specified medical appliances and less than 1/18 of the cost of the common patient monitoring system affordable in markets. As a part of our prototype, heart rate sensor is used over Stethoscopes or Electrocardiogram to respect its accuracy and the user friendliness. In addition to that it is linked with the internet and is serviceable by anyone anywhere.

Recommendations and Future plans

Due to ISEF Competition's short period, we would like to give our recommendations for any further progress to this project:

• In addition to our enormous effort in order to make the device small enough to be handled easily, with the help of the latest technology we recommend that our device would be in a witchlike shape to be portable using the nanotechnology in order to minimize the size of microcontrollers.

• We recommend an integration of monitoring systems into clinical information systems. Such integration reduces multiple redundancy levels in healthcare delivery, decreasing the medical errors from multiple people entering the same data.

• Recently, there is a programmed sensor that can measure diabetes - continuous glucose monitoring (CGM) system - without extracting blood sample from the body so, we hope for this invention to be added to provide a comprehensive information that offers greater insight into clinical patterns and trends, enabling clinicians to make better decisions about patients' conditions.

• It's preferred to program a board with a connection to a AI Software, its function is to show the exact Diagnosis medicament name needed for patient status.

Data Ownership and Governance

Before proposing a linkage, it must be clear that each of the respective data sources allows for the scope of the proposed work. All data owners or key stakeholders need to be contacted, and existing data-governance processes need to be understood. Rules regarding consent and any existing regulatory requirements for the data need to be clarified. The importance of these issues should not be underestimated, and the logistical, administrative, and often legal hurdles need to be anticipated and built into the cost and the timeline for the project.

Cost

Several financial considerations need to be incorporated into project planning and execution for any linkage project. First, obtaining a copy of the data or purchasing a user license often comes with a hefty price. Second, the information technology (IT) systems and technical platform requirements are often significant and can be expensive to build and maintain. However, other options may be available besides building a system de novo. Renting infrastructure from another research partner or computing environment, or utilizing a third-party vendor for data linkage, may provide cost savings and synergy at an organizational level. Lastly, the technical personnel required for linking and maintaining data are often in high demand in this era of “big data” and may require high salaries for recruitment and retention.

Technical Environment and Security

A secure and well-performing computing platform represents the operational backbone for conducting innovative research using large datasets. There is a fine balance between security and usability, and system performance directly influences the size and complexity of the data that can be managed and linked. Careful planning and building collaborations with teams with technical resources help decrease the cost/benefit ratio, while allowing growth in the future, a well-performing and secure research environment builds a foundation of trust with research partners and data suppliers.

Appropriateness and Feasibility of the Project

Each data source needs to be considered carefully regarding its adequacy for the specific linkage endeavor. First and foremost, the *quality* and *discriminatory power* of each of the available linkage identifiers need to be scrutinized. Second, researchers must be confident that there is enough overlap in the two populations to merit the effort. Every linkage will result in a set of matches that will serve as a select, nonrandom, and perhaps not completely representative sample of the two disparate underlying populations. It is important that the extent and representativeness of this potential overlap be considered before beginning the process. Finally, it must be evident that the linkage will result in an enriched dataset that includes additional data elements made possible only through the linkage. Each of these considerations must be weighed together as a whole to determine whether the linkage will provide an adequate return on the investment in time and resources.

Conclusion

Artificial intelligence is a key component of the future of healthcare. Indeed, the era of the AI doctor seems to be unavoidable. Today, we see AI in hospitals helping clinicians identify medical risks; predict when to provide targeted, life-saving interventions; form treatment plans for patients with rare diseases; and deliver precision medicine. At the simplest level, AI chatbots at home can help patients keep on top of care plans. AI applications that remind seniors when to take medication, about doctors' appointments, and even when to eat, can help in removing the anxiety and confusion that many of them face. We have proved this importance of AI in our project. After conducting research and testing our project, we have reached the results expected at the beginning. The first result that matches the hypnosis is the fact that we can reduce the effects on the health care of elderly people and sustain a safe environment for good health. One of the issues elderly people face is the scarcity of health insurance and no doctors near to them to care about them. At the end of our project, we found out that our hypothesis about communication, IOT, and AI, were all true and applicable with high efficiency and expected results as mentioned in the results section. We have been depending on AI for predicting the patient's possible diseases and connecting him with a doctor using back end development. We have determined the number of elderly patients that can fall in any health attack and connected them to doctors who are specialized in the field of the patient disease.

Resources

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